A Novel Approach to Ransomware: Designing a Safe Zone System

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***Abstract*—It is vital to protect the files and data for institutions, organizations and individuals. For this, people and institutions develop methods to provide both physical and digital security, and they spend thousands of dollars over the years for measures. Ransomware is undoubtedly one of the most frequently encountered malicious software that harms people and organizations. These viruses are used to encrypt all important data on the infected computer. Ransomware usually targets file types such as PDF, Word, Excel that can store important documents, bills, financial data, and makes these documents inaccessible by encrypting them. The restoration of these encrypted files is only possible with the keyword in the hand of the person who wrote the virus. This key word is given by an aggressive fee, which means ransom against documents. These fees are usually very high and there is no guarantee that the attacker will not want more fee.**

In this paper, it is aimed to find solutions to these problems and to minimize possible harm. In order to do so, an area called safezone has been created and prevented from being replaced by other software by moving important documents to this zone.

Keywords — information security; ransomware; secure storage; cloud security; computer security

I. INTRODUCTION

In today's information age, in parallel with the fast development of technology, computer and internet usage became more general. This generality caused many works to be done in electronic environment. This situation provides fast growing of data in electronic environment and this spread bring out the necessity of safety collection of these data. Yet, today there are several factors which are threatening security of data. In cyber world, computer users face with threats like worms, phishing, spywares and viruses. In the recent days, one of these malware, which are kind of cryptovirology, occurred and this malware plague to the system, always block the access, lock the victim's data until getting ransom from the victims. These malware, let attackers to get the files in victim's computer and then request ransom to give the original forms of files back.

Ransomware can spread by harmful e-mail attaches, viruses application, infected external devices, and the web sites which have overstepped security systems. In a typical ransomware attack, the attacker can attack systems which are in danger and have vulnerabilities by searching them. Vulnerabilities of the system attacked by a trojan or a worm before can be defined, and it can be attacked easily so the files in the system can be locked.

In this study, a secure file zone has been created in system or computer to protect files against malicious software and attacks, and important documents have been moved to this area for preventing the files from being changed by other software. Software which is running in the operating system has limited access to the file system, and access control is provided only by the user. The developed safezone system, protects the files it contains against ransomware or other malicious software and guarantees this protection with integrity check [1-4].

II. RELATED WORKS

Many studies have been published against ransomware which is a current threat in the literature. In this section, some summaries are given from some of the studies examined as a source of inspiration for this study.

Chan J. and others have some works on Android platform. They have collected 2.721 ransomware example especially from current Android ransomware. They have charactarized data systematically as timeline and malicious features and in some aspects based on these examples. Additionally, its needed customized anti-mobil-ransomware analysis because current antivirus tools are weak about detection. To detect ransomware that locks the data and force users, they have proposed a new and real time detection system which is called as RansomProper. RansomProper can understand whether file lock process performing by users or not, as analyzing UI widgets and coordinates of finger moves of users. Experimental results of works have showed that RansomProper can detect ransomware with an acceptable and high-precision working time performance [5].

Scaife N. and others have presented an early detection system named CryptoDrop. This system warns user during

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a suspicious file process. They have tried to block ransomware by following user's data and normal process of the software. Also, they have tested CyrptoDrop against 492 real world ransomware samples [6].

Zahra S. and others have reviewed growing rate of ransomware which is increasing in last 3 years, the most threatening ransomware attacks for IoT and their statistics. As the conclusion of their works, this growing rate results show that Cryptowall and ransomware numbers are rapidly increasing, therefore, it shows that these ransomware families are a potential threat for IoT [7].

Kanwal M. and others, in their works, have developed an application about static anlayzes of Android ransomware. The application, which was developed with usage of Firebase database on Android devices, is looking for the keywords like ransom, illegal etc. to detect that applications are dangerous or not for users [8].

Vinayakumar R. and others investigated detection and classification of ransomware. They made several tests about web and constructions to choose the best architecture in multi-level probes (MLP). As a result, MLP have showed a better detection and classification performance for ransomware than the other classical machine learning classificators [9].

Kumar S. and others have created a few controlled machine learning classificator by using several malicious and profitable data sets which are created by labeled samples of web traffic features. The main point of the work has identified as Android based malware. Proposed model detected known and unknown threats by 99.4% rate. Also, they have claimed that the model they have proposed, can be integrated with traditional attack detection systems [10].

Yang T. and others, in their works, have provided detection and analysis of ransomware on Android. In that work, malicious software which threats mobile security vulnerabilities has been analyzed and developed automatic analyse approach has been recommended for detection of mobile malicious software detection [11].

Gallegos P. and others have used social engineering method as an attack vector for detection of Ransomware. In their works ransomware anatomy has been analyzed deeply and in this analysis thousands of computers have endangered, they have showed that how they combined manipulatin technology with social engineering. Also, in their works, they have found a base to search about ransomware attacks and other malware on the Internet of Things (IoT) and Software Defined Networking (SDN) [12].

Moore C., in his work, has focused on detection of ransomware by using honeypot techniques.In the work, it was researched that if a bait folder can be created or not for changes. To realize these, File Scanning System of Microsoft File Server Resource Manager and EventSentry to process Windows security logs were used [13].

III. RANSOMWARE SOFTWARE AND MEASURES TO BE TAKEN

In cyber world computer users face with many threats like spywares, worms, phishing attacks, viruses etc. Ransomware let attackers to get the files in victim's computer, lock the files, and than request ransom to give the original forms of files back. In other words ransomware is part of some malware which oversteps the security, sneaks in to the victim's computer and lock all files.

If the ransomware is infected, the attacker keeps the files locked unless the victim accepts to pay the ransom. In a typical ransomware attack, the attacker enters a computer that is in danger by looking for the vulnerabilities of the system. Since this system was weaken by a Worm or Trojan, an attacker can easily get access to the weak system.Then it looks for important file types with extensions such as .txt, .doc, .rft, .ppt, .chm, .cpp, .asm, .db, .db1, .dbx, .cgi, .dsw.If he attacker file extentions like ".gzip, .zip, .jpg, .key, .mdb, .pgp, .pdf", known by victims it is essential for them so these files are locked and impossible to access for the victims. Then the attacker sends an e-mail or pop-up to the victim he/she asks for a ransom that needs an encryption key to open the locked files.

Generally ransomware can spread by harmful e-mails which are sent to users or people. There is no need to run any malware by accessing the system in this kind of attack which is infecting the system by taking advantage of human vulnerability. The transmission of this malicious software is usually via mail. During the infection stage, there is usually an email with a bill issued in the name of the user. When the user sees this bill, which is quite high, it clicks or downloads the bill sent as an attachment or link by panic, and then the malware is infected to the system. In the early versions of ransomware, which were dangerous and rapidly spreading, files could be recovered, even if they were locked, because of some simple mistakes. But when malicious developers relalized this situtation, they prevented this by developing different methods.

Today, the most common types of ransomware are considered as TeslaCrypt, Cryptolocker, TeslaCrypt, Cryptowall, AxCrypt. Ransomware, as a kind of Trojan virus, targets computers which are running the Windows Operating System. Ransomware can spread like a normal file that using social engineering. This malicious software uses a RSA public key to encrypt a private key recorded in the control server to encrypt files in the local network drive in specific formats [3, 4]. This situation is shown in Figure 1.

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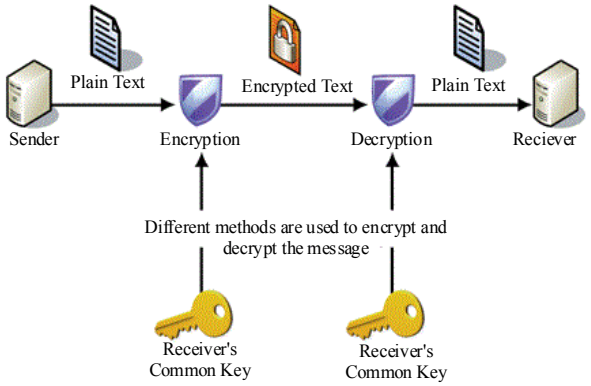

Fig 1. Structure of Ransomware

IV. DEVELOPED APPLICATION

The biggest damage people are faced with during cyber attacks: the loss of files, including data, pictures and documents. Ransomware, which is infected by e-mail or web pop-ups, lock people's data and threat to destroy it if the ransom is not paid. The best protection against this attack is to back up all the information and files on the device in a completely different system during certain periods. To do this, a good place and an offline external hard disk are needed. Such a solution means that even if we're exposed to an attack, we wouldn't exposed to information loss by cyber attackers. This is a solution that requires a certain cost depending on the situation and needs. The aim of this work is to create a safe area for files. This software, which is developed and planned to be run on cloud-based servers in the future, is designed to limit the access to the file system by the operating system software and to give the access control to the user. To do this, the structure of the operating system has been examined deeply and necessary libraries and technologies have been determined and included in the project.

The software keeps all the files in a single file by compressing them for creating a safe area. The name of this file is defined as "safezone.safe" and it is actually a ZIP file and it is contained within the folder. The software manages the contents of this file (adding a new file to the ZIP file, deleting and modifying the file) (ZipManager). To prevent the contents of this file from being changed by other sources, it keeps the file open in non-stop write mode and operating system prevents the file from being changed. This feature is the basis of the secure area (safezone). The software also has an advanced logging system. This system logs all events in the safe zone. Additionally, it starts to watch the main folder of every file added to the safe zone and adds the changes in that folder to the log (FileWatcher). The log file is "safezone.log". In the software, there is also a mechanism to check if the files in created are in danger or not. This integrity control mechanism keeps the MD5 summaries of the files in its own SQLite database called "File.db" and checks integrity for any changes that occur in the file "safezone.safe" according to the information from FileWatcher. So, it compares the initial MD5 summs of the files with the new MD5 summs. Thus, the integrity principle, which is one of the basic criteria of information security, is provided by this hash system.

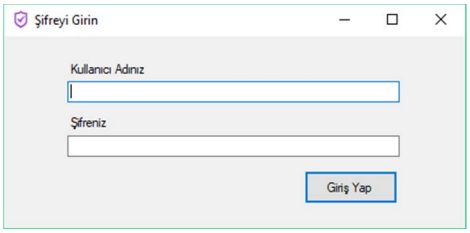

Fig 2. Safezone system login interface

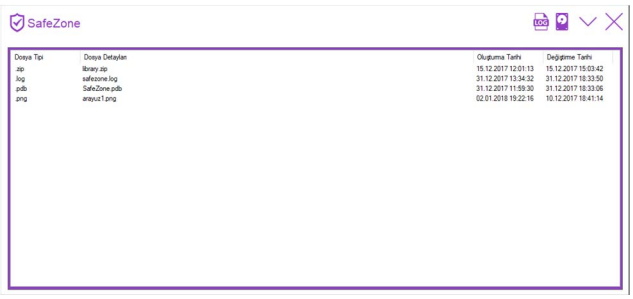

Fig 3. Safezone system file list interface

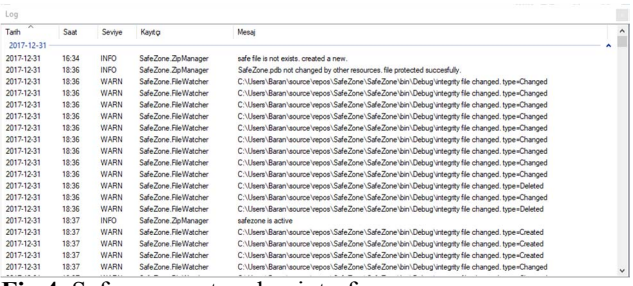

Fig 4. Safezone system log interface

A login screen is shown in Figure 2. in which authorized users of the software can login. With this screen, only previously authorized users will be allowed to login to the system. Figure 3 shows a list of the files that have been imported into the safe zone. This ensures the confidentiality of the safe zone files, which can only be seen by authorized people. Figure 4 shows a screen of the system log which records each event performed on the software. This log screen will record every activity of authorized people on the system.

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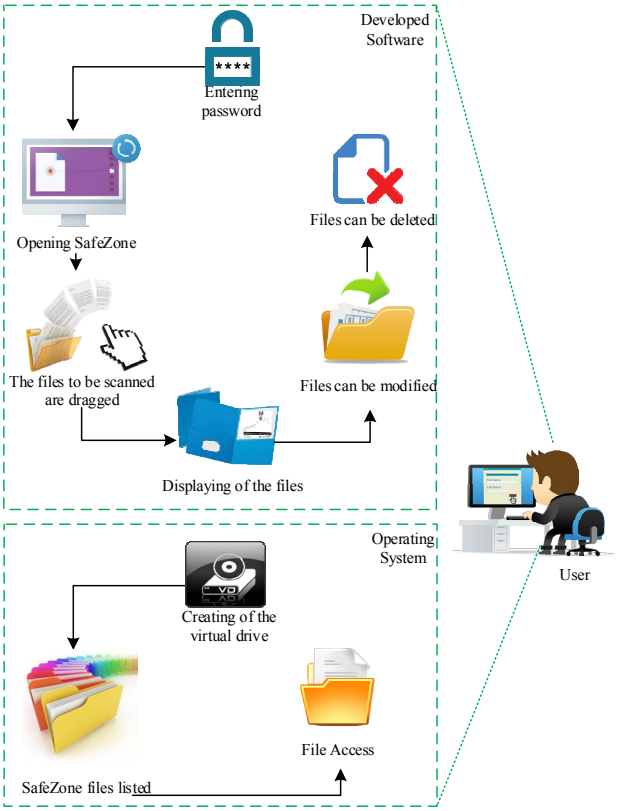

Fig 5. User-based logical scheme of Safezone software

Figure 5 shows the user-based logical scheme of the software. As we can see, the first time the software is opened, a blank zip file called "safezone.safe" is created. The user drags and drops the files, which he/she wants to keep safe, into this zip file with the order of critical importance. Thus, all files are hidden in a single file. From now on, the software will ensure that only one file is safe, and that all files in it are secure. As soon the software run it opens safezone.safe zip file in write mode and never turn off write mode. The operating system allows a file to remain open in write mode only by one source. So other software can not overwrite this file.

The software is automatically run on computer startup and it can not be turned off by the user or another sofware from the task manager. The basis of the software is safezone.safe file. It comes with KeepAlive.exe software. When safezone.safe runs it runs KeepAlive.exe too. When KeepAlive.exe closes, Safezone.exe comes back in milliseconds. When Safezone.exe is closed, KeepAlive.exe software will run Safezone.exe in milliseconds this time. Thus, Safezone.exe software always runs.

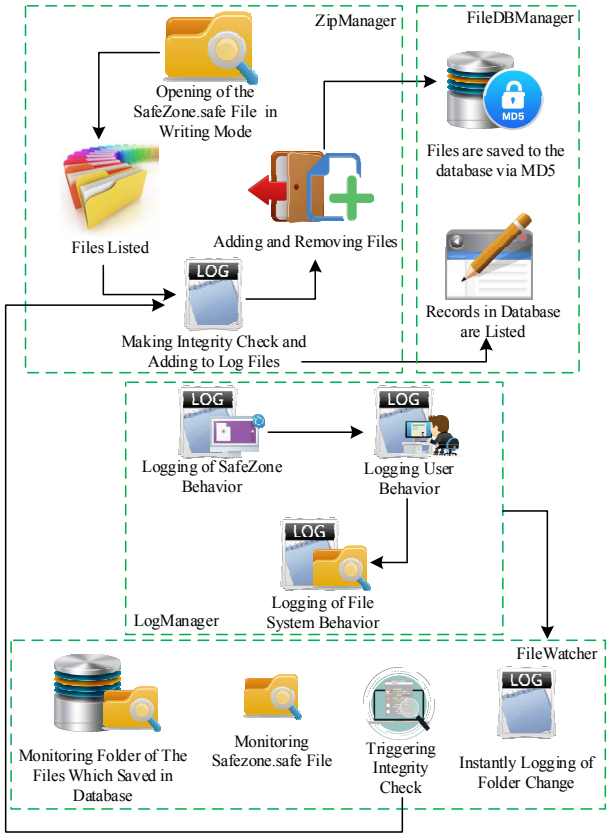

Fig 6. System-based logical scheme of Safezone software

The system-based logical scheme of the software is given in Figure 6. As mentioned above, every file added to Safezone.safe zip file is saved in File.db SQLite database along with file name of MD5 hash. MD5 Hash information is the summary information of the file, and even the smallest change in the contents of the file will completely change the Hash information. At every startup of the program and any changes made to the SafeZone.safe file, it calculates the Hash information of each file in the safezone.safe zip file and compares this hash information with the stored hash information of the related file in the File.db SQLite database. If this information is equal, the integrity of the file is achieved successfully. If it is not equal, file integrity is corrupted. In both cases, it adds the status records to the log file.

V. RESULTS AND EVALUATION

Today, because of the carelessness and lack of interest of the people or the employees of the institution, many institutions may encounter with ransom viruses and they may come into moral and financial harm. In addition, ransom viruses cause data loss. The solution of the files seized in this kind of attack which is impossible to recover is only possible with the randomly generated key in the

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attacker's hand. Every cyber attack with ransomware, causes people to lose reputation as well as material damage in institutions. Many measures have been taken to prevent these losses. In this study, against the effects of ransomware, safe zone solution has been dealt with. The working logic of ransomviruses is to overwrite the files and lock their contents. In this study, it is aimed to prevent the files from being changed or overwritten by other software & viruses by taking user selected files into the secure zone. At the same time, a mechanism has been developed that checks whether integrity has been achieved for certain periods by performing integrity checks.

It's aimed that increasing usability with a basic interface which does not make the user feel the complicated operations in the background. In addition, a system has been created that watches the file movements on the disk in the background of the system and keeps log records. This allows the user to be aware of file movements in the system. To prevent files from being overwritten, the operating system has used a single writing permission to write to the file. First, a file was created in which the authorized user in this software can "write" by ensuring that the system executed and the permission to write are obtained. It is aimed to develop both a software and a hardware system that can provide better performance in following works. In addition to this, we aim to realize some works about filtering and checking sofware which can make "write" process on a disk. So we can develop a module that can provide the security of the disks at a high level. Thus the system to be developed will also serve as a cloud service to provide enterprise information security.

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